



福昕PDF编辑器

• 永久 • 轻巧 • 自由

升级会员

批量购买



永久使用

无限制使用次数



极速轻巧

超低资源占用，告别卡顿慢



自由编辑

享受Word一样的编辑自由



扫一扫，关注公众号

3

Creating Test Protocols

Learning Objectives

After completing this lab, you should be able to:

- Provide enough information about the test features of your design that DFTC can create a test protocol.
- Verify that you have told DFTC what it needs to know
- Verify that your protocol and design are compatible
- Verify that the design has been compiled to the correct library and that all flip-flops have been replaced with scan flip-flops
- Modify a test protocol to include an initialization sequence for applying an internal test mode signal



Lab Duration:
60 minutes

Lab 3

Part A Overview

The files for **Part A** of this lab are located in the directory **lab3a_protocol**.

Directory Structure

lab3a_protocol	Current working directory
analyzed	Intermediate acs_read_hdl files
logs	Session log files
unmapped	Unmapped protocol
mapped	Gate-level netlist
reports	DFTC reports
scripts	Constraint and run scripts
../ref/rtl/RISC_CORE/vhdl	Design files
../ref/db	Library files

If you need help...

```
dc_shell> help prev*
dc_shell> printvar *scan*
dc_shell> preview_dft -help
dc_shell> man compile
```

When working interactively, you do not need to type the entire command or option.

```
dc_shell> read_test_p -v
dc_shell> all_inputs -cl
```

Answers & Solutions

This lab guide contains answers and solutions to all questions. If you need some help with answering a question, consult the back portion of this lab for help.

Instructions

During this lab, you will write a script to create and verify a test protocol and explore a `dc_shell` log file.

Task 1. Create a test protocol with a script

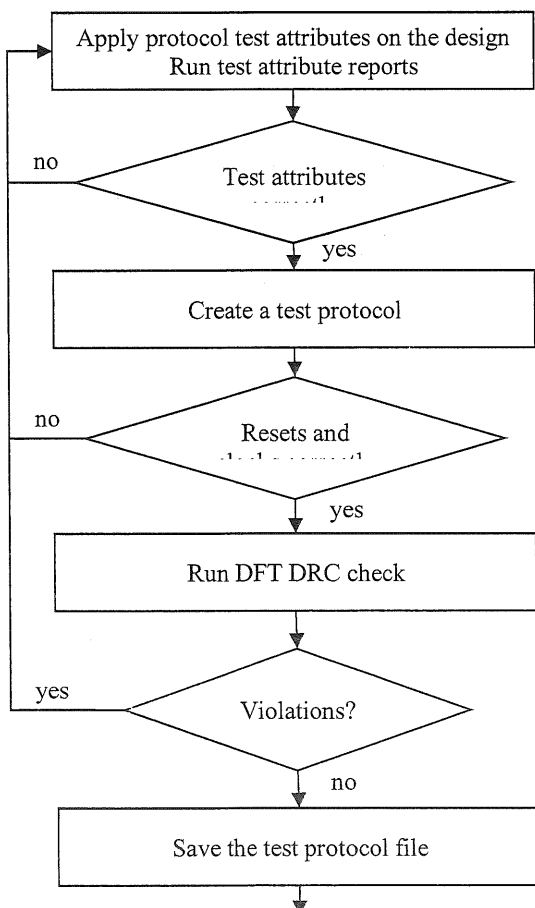
Utilize the flowchart below and the protocol test attributes specification to its right as a guide to write and verify a DFTC script to create a test protocol.

1. Change directories to the project directory `lab3_protocol`.

Look at the “dot” files in that directory.

```
UNIX% cd lab3a_protocol
```

Test Protocol Definition Flow



Protocol Test Attributes	
Test Clock Port	Clk
Test Clock Active Time	45 ns
Test Clock Inactive Time	55 ns
Test Mode Port	TEST_MODE
Reset port	Reset
	(asynchronous active low)

Lab 3

- Open the script called `scripts/unmapped.tcl`.
The tasks are defined in the flow chart above; the values for the attributes in the chart beside it.
- Your task is to write the DFTC commands to perform the tasks. For this lab, set the values for the protocol test attributes in the script.
- Execute your script by running the **unmapped.tcl** script in batch mode.
Create a log file for the run:

```
unix% dc_shell -f scripts/unmapped.tcl | \
tee logs/run_unmapped.log
```

- Determine if there were any problems in the unmapped flow by looking at `logs/run_unmapped.log`.

Question 1. What command is used to see which DFT signals have been applied to the design?

set app var
设置系统变量

report_dft_signal

- Question 2.**
- What value is the TEST_MODE port set to?
 - What is the name of the test clock port?
 - What is the period of the test clock?
 - When does the test clock go high?
 - When does the test clock go low?
 - What is the name of the reset port?
 - What type of reset has been assumed?

suppress message
忽略不关心或没问题的
warning

- 1
- CLK
- 100ns
- 45ns
- 55ns
- Reset
- asynchronous active low

```
dc_shell> report_dft_signal
.....
Report : DFT signals
Design : RISC_CORE
Version: H-2013.03-SP5
Date : Wed Sep 5 14:35:36 2018
.....
TEST MODE: all dft
VIEW : Specification
No DFT signals defined in this mode.
.....
Report : DFT signals
Design : RISC_CORE
Version: H-2013.03-SP5
Date : Wed Sep 5 14:35:36 2018
.....
TEST MODE: all dft
VIEW : Existing DFT
Port SignalType Active Hookup Timing Usage
-----
TEST_MODE TestMode 1 - -
CLK ScanMasterClock 1 - P 100.0 R 45.0 F 55.0
CLK MasterClock 1 - P 100.0 R 45.0 F 55.0
Reset Reset 0 - P 100.0 R 55.0 F 45.0
P0W[0] Constant - -
dc_shell>
```

Question 3. Look at the results of running the `dft_drc` command, are there any errors or warnings?

```
dc_shell> create_test_protocol
In mode: all_dft...
Information: Starting test protocol creation. (TEST-219)
...reading user specified clock signals...
Information: Identified system/test clock port CLK (45.0,55.0). (TEST-265)
...reading user specified asynchronous signals...
Information: Identified active low asynchronous control port Reset. (TEST-266)
.....
dc_shell> dft_drc
In mode: all_dft...
Pre-DFT DRC enabled
Information: Starting test design rule checking. (TEST-222)
Loading test protocol
...basic checks...
...basic sequential cell checks...
...checking for scan equivalents...
...checking vector rules...
...checking pre-dft rules...
.....
DRC Report
Total violations: 0
.....
Test Design rule checking did not find violations
.....
Sequential Cell Report
0 out of 261 sequential cells have violations
.....
```

- Exit from DFTC.

Lab 3-4

Synopsys DFT

① 报告 violation 怎么找
有问题的 cell
② 为何不是按寄存器报告的

Part B Overview

All files for Part B of this lab except for designs and libraries are located in the directory **lab3b_init**.

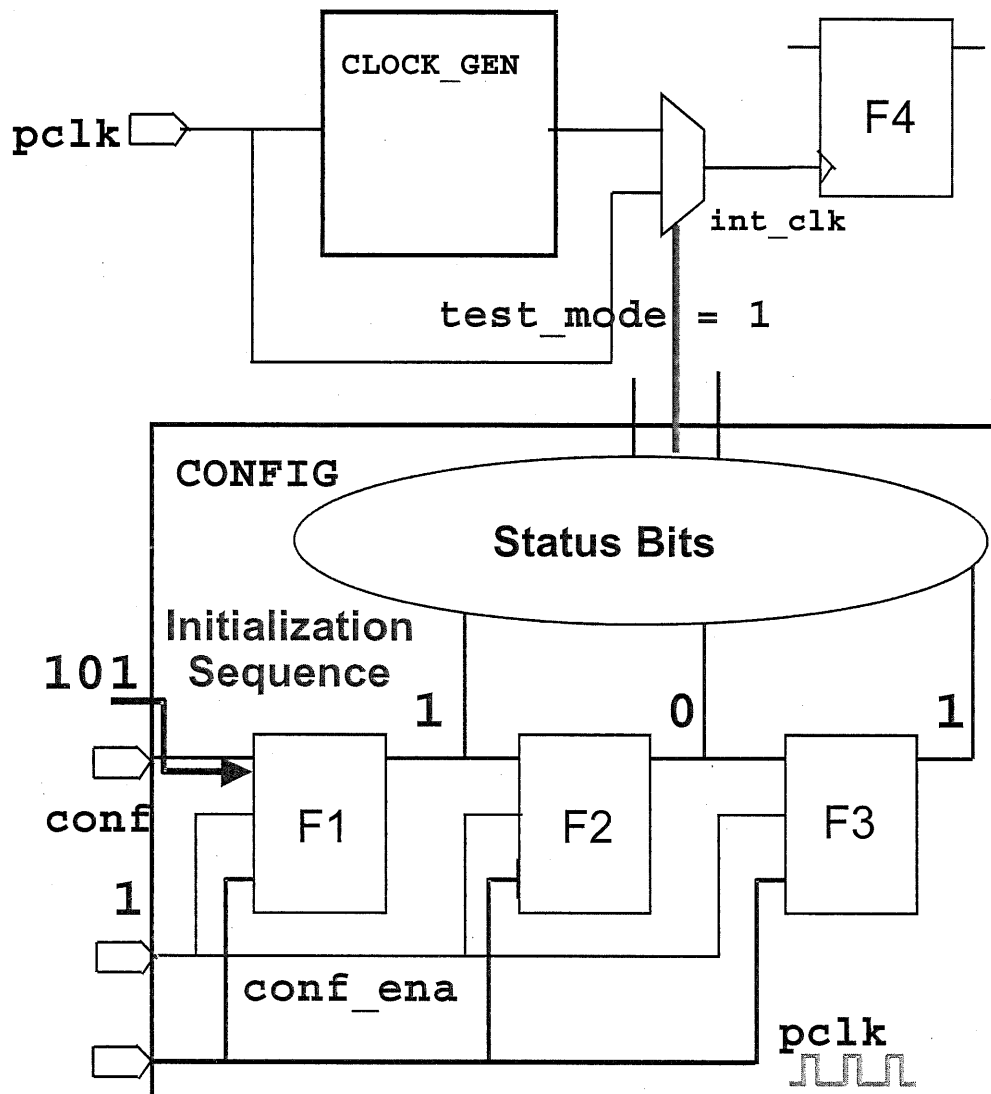
Directory Structure

lab3b_init	Current working directory
analyzed	Intermediate acs_read_hdl files
logs	Session log files
unmapped	Unmapped protocol
mapped	Gate-level netlist
mapped_scan	Gate-level scan design netlist
reports	DFTC reports
tmax	Files for downstream tools
scripts	Constraint and run scripts
../rtl/ORCA_init/vhdl	Design files
../libs	Library files

Lab 3

Design Introduction

The design used in this task requires a custom test initialization protocol. You obtained the following schematic from the ORCA design documentation that describes the process for asserting the internal test_mode signal.



Instructions

Since DFTC cannot infer a test protocol of this type you must develop one for this design. Answer the following questions in preparation for doing the subsequent lab tasks.

Task 1. Internally Generated test_mode Signal Using the CONFIG Block

Question 4. What value must be placed on the conf_ena signal to initialize the configuration register flip-flops F1, F2 and F3?

1

Question 5. How many cycles must you pulse pclk before test_mode is asserted?

3

Question 6. What values should conf_ena and pclk have the cycle after test_mode is asserted?

both 0

Question 7. Write in STIL format the Vector statements that would go into the "test_setup"?

```

. . .
MacroDefs {
  "test_setup"
  {
. . .
1078      V {"conf_ena" = 1 ; "conf" = 1 ; "pclk" = P ;}
1079      V {"conf_ena" = 1 ; "conf" = 0 ; "pclk" = P ;}
1080      V {"conf_ena" = 1 ; "conf" = 1 ; "pclk" = P ;}
1081      V {"conf_ena" = 0 ; "pclk" = 0 ;}
-----
. . .
  }

```

You will use this information later when you develop the initialization protocol for ORCA.

During this lab, you will develop a custom test initialization protocol for a design that has an internal test_mode signal.

Lab 3

Task 2. Read a Mapped Design and Create a Test Protocol

1. Change directories to the project directory `lab3b_init`.

```
UNIX% cd lab3b_init
```

2. Invoke DFTC.

```
dc_shell
```

3. Create a shortcut to make running the scripts easier.

```
alias s "source -echo -verbose"
```

4. A mapped design has been prepared for you in the interest of saving time during lab (Reading the RTL and compiling).

Examine the contents of the script that reads the saved ddc file for the ORCA design and then source the script to read in ORCA.

```
exec cat scripts/lread_design.tcl  
  
s scripts/lread_design.tcl
```

Question 8. Where is the design read from? Is the design RTL or mapped gates?

from
mapped/ORCA.ddc

mapped gates

Before you create the starting test protocol you must specify all your clocks, resets, constants, etc. The clocks and the resets are already defined in the `2create_test_protocol.tcl` script.

Question 9. After the test initialization asserts the internal `test_mode` signal what must you do to keep the ORCA design in that mode throughout the rest of the scan testing?

hold conf_ena at 0

```
dc_shell> set_dft_signal -view existing_dft -type Constant -active_state 0 -port conf_ena
```

```
dc_shell> set_dft_signal -view existing_dft -type Constant -port conf_ena -active 0
```

5. Add the appropriate command to the `2create_test_protocol.tcl` script.

```

DRC Report
Total violations: 4788
-----
13 MODELING VIOLATIONS
  13 Cell has unknown model violations (TEST-451)
432 TOPOLOGY VIOLATIONS
  432 Improperly driven three-state net violations (TEST-115)
4325 PNR-DFT VIOLATIONS
  2946 Uncontrollable clock input of flip-flop violations (D1)
  26 DFF set/reset line not controlled violations (D2)
  1246 DFF set/reset line not controlled violations (D3)
  6 Clock feeding multiple clock/set/reset inputs violations (D12)
  1 Clock path affected by clock captured by clock in trailing edge clock_port violation (D16)
  18 Clock_port not capable of capturing data violations (D17)
Warning: Violations occurred during test design rule checking. (TEST-124)

```

Lab 3

6. Create the test protocol for ORCA by running the script.

```
s scripts/2create_test_protocol.tcl
```

Review the results of the `dft_drc` run on the ORCA RTL design.

Question 10. Are there few or many testability problems reported?

many...

Task 3. Create an Initialization Sequence

Developing the initialization test protocol from scratch is not practical for most engineers. A better method is to modify the protocol files that DFTC or TetraMAX create for you.

1. Save out the current test protocol for the design.

```
write_test_protocol -o orca_mapped.spf
vi orca_mapped.spf
```

2. Add the required Vector statements ("`V { }`") to the `orca_mapped.spf`.

Question 11. Since you only changed the "`test_setup`" section of the STIL test protocol, which command should you use to read it back into DFTC?

```
dc_shell> read_test_protocol -section test_setup orca_mapped.spf
```

3. Verify that your initialization vectors are correct by reading in your custom protocol and rerunning `dft_drc`.

Question 12. What happens when you first try to read `orca_mapped.spf` back into DFTC?

```

Information: Initialization protocol will be applied to all_dft mode
Warning: a protocol already exists in mode 'all_dft.' Please use remove_test_protocol command to
delete the existing test protocol. (TEST-1402)

```

4. Solve the problem by removing the previous protocol and reading in your custom initialization.

```
remove_test_protocol
read_test_protocol -section test_setup \
                  orca_mapped.spf
```

Note: The "`test_setup`" or initialization section is only part of a complete test protocol.

Creating Test Protocol:
Synopsys DFT Compiler

```

dc_shell> remove_test_protocol
... deleting protocol from the mode 'all_dft' ...
dc_shell> read_test_protocol -section test_setup orca_mapped.spf
Information: Initialization protocol will be applied to all_dft mode
... Reading /home/njupt/Lsh/Zsynopsys_LAB/dft/dftc1_2013.03/DFTC1_2013.03/lab3b_init/orca_mapped.spf ...
... STIL version 1.0 ( Design 2005 CTL P2001.10) ...
... Reading MacroDefs "test_setup" ...
...
Read protocol for mode all_dft

```

Lab 3-9

Lab 3

5. Re-create the rest of the test protocol before attempting to run `dft_drc`.

```
create_test_protocol
dft_drc
```

Question 13. How do the results of `dft_drc` compare to your first run?

MODELING VIOLATIONS & TOPOLOGY VIOLATIONS no changes;
fewer PRE-DFT VIOLATIONS;
new OTHER VIOLATIONS: TEST-504, TEST-505.

Question 14. Explain why the 3 configuration registers will not be placed in the scan chains?

To prevent changing the working state of the chip due to register actions during testing

Congratulations! You have completed this lab.

to viola

```
-----
DRC Report
Total violations: 494
-----

13 MODELING VIOLATIONS
  13 Cell has unknown model violations (TEST-451)

432 TOPOLOGY VIOLATIONS
  432 Improperly driven three-state net violations (TEST-115)

46 PRE-DFT VIOLATIONS
  6 Clock feeding multiple clock/set/reset inputs violations (D12)
  20 Data path affected by clock captured by clock in trailing edge clock_port violations (D14)
  20 Clock_port not capable of capturing data violations (D17)

3 OTHER VIOLATIONS
  1 Cell is constant 0 violation (TEST-504)
  2 Cell is constant 1 violations (TEST-505)

Warning: Violations occurred during test design rule checking. (TEST-124)
-----
Sequential Cell Report
35 out of 3002 sequential cells have violations
-----

SEQUENTIAL CELLS WITH VIOLATIONS
* 32 cells have test design rule violations
* 1 cell has constant 0 value
* 2 cells have constant 1 value
SEQUENTIAL CELLS WITHOUT VIOLATIONS
* 2967 cells are valid scan cells

Information: Test design rule checking completed. (TEST-123)
1
```